DE–681

Sub. Code
101

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Perform the indicated base conversions:
   (a) 54_8 to base 5
   (b) 312_4 to base 7.

2. Write the Boolean expression for a four-input NAND gate.

3. Construct the operation XOR from the basic Boolean operations AND, OR, and NOT.

4. Design an 1-to-8 demultiplexer.

5. How can you form the negation of an integer in two's complement representation?

6. Assume numbers are represented in 8-bit two's complement representation. Show the calculation of the following: (a) –6 + 13 (b) 6 – 13.

7. What is the difference between DRAM and SRAM in terms of characteristics such as speed, size, and cost?

8. What is the difference between memory-mapped I/O and isolated I/O?

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is the difference between cannonical form and stardard form? Which form is preferable when implementing a Boolean function with gates? Which form is obtained when reading a function from a truth table?

10. Obtain the simplified expressions in product of sums:

$F(w, x, y, z) = \pi (1, 3, 5, 7, 13, 15)$

11. Implement full adder circuit with multiplexers.

12. Implement a full subtractor with two half subtractors and an OR gate.

13. Design a BCD counter with JK flip-flops.

14. Suggest reasons why RAMs traditionally have been organized as only 1 bit per chip whereas ROMs are usually organized with multiple bits per chip. Consider a dynamic RAM that must be given a refresh cycle 64 times per ms. Each refresh operation requires 150 ns; a memory cycle requires 250 ns. Whet percentage of the memory's total operating time must be given to refreshes?

15. (a) When a device interrupt occurs, how does the processor determine which device issued the interrupt?

    (b) When a DMA module takes control of a bus, and while it retains control of the bus, what does the processor do?

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DE–682

Sub. Code
102

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

C AND DATA STRUCTURES

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various operators in C.

2. Illustrate working style of the nested 'If- Else' structure.

3. Differentiate the Call by Value from Call by Reference.

4. Write a note on C processor.

5. What is dynamic memory allocation? Explain with examples.

6. Illustrate the differences between Structures and Arrays.

7. Discuss the circular queue with an example.

8. Explain the singly linked list.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Illustrate the different types of looping structures.

10. Write a program using function in C language to multiply the matrices A and B with orders 2×3 and 3×4.

11. How do you pass pointer variables to a function? Illustrate.

12. What is file pointer? Write a program to create a file and store data.

13. Describe the operations on Stack.

14. Discuss the Doubly linked list with examples.

15. Write an algorithm for searching an element in a binary search tree. Give an example.

DE–683

Sub. Code
103

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is Authentication? Explain how it is differ from Authorization?

2. What is meant by Enhanced Data Model? Explain.

3. Discuss how Multivalued Dependencies are eliminated in 1NF.

4. What are the challenges and issues in Data Base Security Design?

5. Explain how Database Security is achieved using SQL?

6. Briefly explain the Host and Indicator variables in SQL.

7. Explain Correlated sub queries with comparison Operator.

8. Write a note on Ordered Indexing and Hash Indexing.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Briefly explain about various Relational Algebra Expressions with example.

10. Diagrammatically illustrate the steps involved in Query Processing.

11. Explain the various methods to improve Data Integrity in SQL.

12. What are Cursors? How cursor attributes are handled in SQL? Explain with programming example.

13. (a) Briefly explain the overview of oracle utilities in data model. (7)

    (b) Explain the various constraints in relational model. (8)

14. (a) Define Normalization. Explain Fourth Normal Form with example. (7)

    (b) Explain Log Based Recovery Techniques in SQL. (8)

15. Briefly explain about executed embedded SQL statement with example.

DE–684

Sub. Code
104

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

First Semester

OPERATING SYSTEMS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a short note on history of Operating Systems.

2. Explain the race condition.

3. Write short notes on Message passing.

4. Explain the Priority scheduling.

5. Discuss the various I/O devices.

6. Give a brief note on the major problems to implement the demand paging.

7. Write a note on Bit-maps.

8. How does an operating system shares a file? Explain.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the Kernel structure of on Operating System.

10. Discuss the Round Rotrin Scheduling technique.

11. Describe the various functions of Interrupt handler.

12. What is meant by device independent software? Discuss.

13. Write a detailed note on memory partitioning.

14. Give a detailed description on deadlocks and its characterization.

15. Discuss in detail about the protection mechanism in File system.

DE–9546

Sub. Code
105

DISTANCE EDUCATION

M.C.A.(S) DEGREE EXAMINATION, MAY 2017.

First Semester

DATA STRUCTURE USING C LAB

(2010 Academic Year onwards)

Time : Three Hours Maximum : 100 Marks

Examiner has to select ONE question and give it to the student by the lot system.

Each subdivision carries 50 marks.

1. (a) Write a C program to convert a given number into words.

   (b) Write C program to implement push and pop operation of stack.

---------- Cut here ----------

2. (a) Write a C program to copy the contents of one file to another file.

   (b) Write a C program to find a given word is palindrome or not.

3. (a) Write a C program to find the roots of a Quadratic equation $ax^2 + bx + c = 0$ using function definition.

   (b) Define selection sort and write a C program to sort a set of elements using selection sort.

---------- Cut here ----------

4. (a) Write a C program to evaluate a given mathematical expression using stack.

   (b) Write a C program to find the factorial of a given number using recursion.

---------- Cut here ----------

5. (a) Names and marks of 10 students in 3 different subjects are given. If a student scores more than 40 marks in each subject, he is declared 'PASS', otherwise 'FAIL'. Write a C program to do the above using structure concept.

   (b) Write a C program to evaluate any post fix mathematical expression using stack.

DE–9547

Sub. Code
106

DISTANCE EDUCATION

M.C.A DEGREE EXAMINATION, MAY 2017.

First Semester

RELATIONAL DATABASE MANAGEMENT SYSTEMS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select any give one question to each candidate from 1st system.

1. A Hospital Information System Maintains Blood Donors Record information in a file. The File contains the following Fields:

   Donor number, Donor name, Donor age, Donor address, Pin code, Place of birth, Blood group (A, B, AB and O).

   Create the Table and Insert 10 Records based on following Queries.

   Write a program in SQL to Display the Donor number, Donor name and address of the donors for the following categories :

   (a) Blood donors having AB blood group.

   (b) Blood donors in the age group between 16 to 25.

   (c) Female donors with 'A' blood group in the age between 18 to 25.

   (d) Blood donors who are living in Chennai only.

   (e) Blood donors whose name begins with S only.

2. A Salary Statement contains the following Fields.

Employee No, Employee Name, Basic Pay, HRA, DA, PF, IT, Gross Pay, Net Pay. Create the Table and Insert 10 Records.

Calculate :

HRA = 5% of Basic Pay

DA = 75% of Basic Pay

PF = 10% of Basic Pay

Gross pay = Basic Pay + DA + HRA

IT = 10% of Gross Pay

Net Pay = Gross Pay — (PF + IT)

Write a program in SQL to prepare Salary Report for 10 Employees.

---------- Cut here ----------

3. Write a program in SQL for Hospital Management System with the following fields :

Patient no, Patient name, Age, Attending Doctor, Patient _ Type (IN/OUT),

Consulting Fee, Blood Test Fee, X-Ray Fee, Other Fee and Total Fee.

Create the Table and Insert 10 Records based on following Queries.

Write the Query in SQL to Display the following conditions :

(a) The patients, who take Blood test and X-Ray

(b) The patients, whose Name ends with R.

(c) The patients, who belong to In-patient category.

(d) Calculate the total fee collected for the entire Day.

(e) The Patients who take the treatment for the age more than 60.

4. An Airline Reservation Database contains the Reservation table and Personal table.

Reservation Table contains following fields:

Flight no, Class, Passenger name, Seat Number.

Personal table contains following fields:

Passenger name, Sex, Age, Marital Status, Nationality

Create Both the Table and Insert 10 Records based on following Queries.

Write a program in ORACLE to Display the following Conditions :

(a) List the Passenger Name, seat number, class and Flight number.

(b) Total number of Married Female Passengers in a particular Flight.

(c) List out the male passengers in the age between 18 to 25 in a Flight.

(d) Display how many Indians are in the Flight?

(e) List the Passenger Name starts with S.

5. Tamil Nadu Electricity Board Billing System contains the following Fields.

Consumer No, Consumer Name, Address, Current Value, Previous Value, Chargeable Unit, Code (D stands for Domestic, C stands for Commercial), Pay Mode (Cheque/Cash)

Calculate the Bill based on following Criteria.

For Code D Rs. 2 Per Unit. (If Chargeable Unit < = 200)

Rs. 3 Per Unit (If Chargeable Unit > 200)

For Code C Rs. 5 Per Unit. (If Chargeable Unit < =200)

Rs. 7 Per Unit (If Chargeable Unit > 200)

Create the Table and Insert 10 Records based on following Queries. Write the Query in SQL to Display the following conditions :

(a) Display the Consumer Name who paid the bill in Pay Mode Cheque

(b) List the Consumer Number who paid the bill More than 2000 Rs.

(c) Find the Consumer Name who paid the Lowest Amount on that Day.

(d) Display the Consumer Number who paid the Bill Code C.

(e) Display the Consumer Name who are Living in Chennai.

6. Create a Table Employee with the following Fields. Emp no, Emp name, Dept no, Doj, Designation

   Insert 10 Records based on following Queries.

   (a) Alter the table employee, make Emp no as Primary key and Dept no as Not Null.

   (b) Increase the column width by 3 for Deptno field.

   (c) Add the new field as Address.

   (d) List all the employees whose name begin with 'V'.

   (e) Display all the employees who have joined before 31-DEC-2013.

---------- Cut here ----------

7. Library Information System maintains the following Fields.

   Roll no, Name, Book No, Book Name, Author Name, Publisher Name, Issue Date, Return Date, Fine students should Return the Book within 10 Days. Otherwise Fine should be collected for each Day Rs. 5

   Insert 10 Records based on following Queries.

   (a) Display the Book Number, Publisher Name only.

   (b) Display Total Amount of Fine Collected.

   (c) Display BPB Publisher Books Only.

   (d) List the students Name those who are Return the Book Within 10 Days.

   (e) List all the Author Name whose name ends with 'R'.

8. Write a PL/SQL Program to Prepare a Mark Statement of a Student using Student Table and Mark Table Using Cursor.

---------- Cut here ----------

9. Write a PL/SQL Block to insert Records into Student Table using Stored Procedure.

---------- Cut here ----------

10. Write a PL/SQL Block to find the Reverse of given String using stored function.

DE–685

Sub. Code
201

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. List the features of object oriented programming.

2. What are the applications of OOP? Explain.

3. Define in-line function? Explain with an example.

4. Explain the function overloading concept with an example.

5. Describe the 'pass by value' and 'pass by reference' with an example.

6. Give notes on virtual function.

7. Explain the file management concept in C++.

8. How will you create a random access file in C++? Illustrate.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the structure of a C++ program with example.

10. What are the different types of inheritance? Explain them with an example.

11. Write a C++ program to multiply two matrices.

12. Explain the parameterised constructor with example.

13. Define pointers? Explain the advantages of using pointers in C++? Give example.

14. Briefly describe the exception handling with an example.

15. Discuss the various file handling functions with an example.

DE–686

Sub. Code
202

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2017.

Second Semester

UNIX AND SHELL PROGRAMMING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What is meant by shell? How do you execute shell scripts? Explain.

2. Explain the directories in Unix.

3. Explain the various looping structures available in shell programming.

4. Explain the various methods of evaluating arguments.

5. Explain the pick and news command with an example.

6. Write short notes on Unix system calls.

7. Explain the method of creating a file in Unix using Vi editor.

8. Write short notes on Tbl and eqn preprocessors.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the Unix system application environment.

10. How do you execute a command without invoking a new process? Explain.

11. How to trap and catch the interrupt in shell programming? Explain with an example.

12. Write a shell program to check whether the given number is prime or not.

13. What do you mean by redirection? Explain the I/O redirection.

14. List and explain any ten Unix commands with an example.

15. Write short notes on :

    (a) MS-Macro package

    (b) Document preparation tools.

DE–687

Sub. Code
203

DISTANCE EDUCATION

M.C.A (S) DEGREE EXAMINATION, MAY 2017.

Second Semester

COMMUNICATION SKILLS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. What are the reading techniques?

2. Explain the importance of communication.

3. Write a note on the art of conversation.

4. How do you plan for an interview?

5. Define and explain the purpose of group discussion.

6. Trace the origin and development of body language.

7. What are the advantages of mock negotiation?

8. Write a note on communication within management.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What are the processes to develop communication skills?

10. What are the various types of the interview?

11. Explain the features of group discussion.

12. How could communication skills for a meeting be made effective?

13. Trace the fundamental uses of negotiation techniques.

14. Distinguish downward and upward communication.

15. Describe written and oral communication.

DE–688

Sub. Code
204

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

Second Semester

COMPUTER GRAPHICS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the various Graphical input devices.

2. Write a note on Graphics system software.

3. Discuss the two-dimensional composite transformation.

4. What is meant by viewports? Explain with a suitable example.

5. Write short notes on Mirror reflection.

6. Discuss the Orthogonal projection.

7. Discuss the line clipping in three dimensional view.

8. Write a note on Command languages.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write and explain an algorithm to draw ellipse.

10. Describe the region filling technique.

11. Explain the Matrix representation of the 2-D transformation.

12. Discuss the polygon clipping algorithm.

13. Explain the composite 3-D transformations.

14. Explain the Hidden surface algorithm.

15. Describe the depth buffer algorithm.

DE–9548

Sub. Code
205

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, MAY 2017.

Second Semester

C++ AND GRAPHICS LAB

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question by the lot system and give it to the student.

1. (a) Write a C++ program to draw a straight line using Bresenham's algorithm. (55)

   (b) Write a C++ program to rotate and scale the two-dimensional object. (35)

   Record (10)

---------- Cut here ----------

2. (a) Write a C++ program to draw a straight line using DDA line drawing algorithm. (50)

   (b) Write a C++ program to draw an ellipse and fill it with any one color. (40)

   Record (10)

---------- Cut here ----------

3. (a) Write a C++ program to draw a circle and ellipse using Bresenham's algorithm. (55)

   (b) Write a program to draw the 3-dimensional object and perform the animation. (35)

   Record (10)

4. (a) Write a C++ program to implement the polygon clipping algorithm using graphics mode functions. (55)

(b) Write a C++ program to draw the non-linear object arc and curve. (35)

Record (10)

---------- Cut here ----------

5. (a) Write a C++ program to draw a polygon and three concentric polygons. (40)

(b) Write a C++ program to manipulate the graphical object using shading technique. (50)

Record (10)

---------- Cut here ----------

6. (a) Develop a C++ program to translate and rotate the two-dimensional object. (45)

(b) Write a C++ program to show the graphical transformation on three dimensional objects. (45)

Record (10)

---------- Cut here ----------

7. (a) Write a program to draw a ball and perform the movement of the ball from one end to another end. (40)

(b) Develop a C++ program to draw a non-linear object polygon. (50)

Record (10)

8. (a) Write a C++ program to draw a circle, divide it into four quadrants and fill up with different colors. (45)

(b) Develop a C++ program to draw cube. Also specify its coordinates. (45)

Record (10)

DE–9549

Sub. Code
206

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION,
MAY 2017.

Second Semester

UNIX AND SHELL PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the examines and given to the candidates by lot system

1. (a) Write a shell program to check whether the given file is readable or writable.

   (b) Write a shell program to determine the given number is armstrong or not.

---------- Cut here ----------

2. (a) Write a menu driven shell program for the following

   (i) List of files

   (ii) Today's date

   (iii) Quit.

   (b) Write a shell program to convert all capital letters in a file to a small case letters.

3. (a) Write a shell program to sum up the following series

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ldots+\frac{1}{10!}.$$

(b) Write a shell program to display the list of last 20 files present in the current directory. Also store this list in a file profile.

---------- Cut here ----------

4. (a) Write a shell program to calculate factorial of a given 'n' number.

(b) Develop a shell program to read *'n'* names and sort, store them in a separate file.

---------- Cut here ----------

5. (a) Write a shell program which accepts the name of a file from the standard input and check the following

(i) File existence

(ii) File readable

(iii) File writ able.

(b) Write a shell program to print the fibonacci series upto 100 numbers.

6. (a) Merge the contents of the files file 1, file 2 and sort them and display the sorted output on the screen page by page.

   (b) Write a shell program to concatenate and copy two strings without using built-in functions.

---------- Cut here ----------

7. (a) Write a shell program to reverse the digits of an integer.

   (b) Write a shell program to enter the sentence in file, search a given word in an existing file and quit.

DE–689

Sub. Code
301

DISTANCE EDUCATION

M.C.A./M.C.A. (LE) DEGREE EXAMINATION, MAY 2017.

Third Semester

DISCRETE MATHEMATICS

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions

1. Give the principal disjunctive and conjuctive normal forms of the expression $\left(\neg P \to R\right) \wedge (Q \leftrightarrow P)$.

2. Using indirect method of proof, derive $P \to \neg S$ from $P \to Q \vee R, Q \to \neg P, S \to \neg R, P$.

3. (a) Show that the set of real numbers in [0, 1] is uncountable set.

   (b) If $A = \{1, 2, 3\}, B = \{2, 3, 4\}$ and $v = \{1,2,3,4,5,6\}$ verify that

   (i) $(A \cup B)' = A' \cap B'$

   (ii) $(A \cap B)' = A' \cup B'$.

4. Prove that $A - (B \cap C) = (A - B) \cup (A - C)$.

5. Use the definition of order to show that $5x^3 - 3x + 4$ is $0(x^3)$.

6. A subgroup H of a group G is normal if and only if $g^{-1}hg \in H$ for every $h \in H$, $g \in G$.

7. Prove that "A connected graph has atleast one spanning tree".

8. Prove that in an acyclic digraph, there exists atleast one source and atleast one sink.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write the following sentences in the closed form or symbolic form

(a) Some people who trust others are rewarded. (3)

(b) If any one is good then John is good. (3)

(c) He is ambitious or no one is ambitious. (3)

(d) Some one is teasing. (3)

(e) It is not true that all roads lead to rome. (3)

10. (a) Express $\sqrt{2}$ is an irrational number using quantifiers. (7)

(b) Let $Q(x)$ be the statement $x < 2$. What is the truth value of the quantification $\forall x\, Q(x)$, where the universe of disclosure consists of all real numbers? (8)

11. (a) Show that the relation $(x, y)R(a,b) \Leftrightarrow x^2 + y^2 = a^2 + b^2$ is an equivalence relation on the plane and describe the equivalence classes. (8)

(b) If R be a relation in the set of integers Z defined by $R = \{(x, y) : x \in Z, y \in Z, x - y \text{ is divisible by } 3\}$. Describe the distinct equivalence classes of R. (7)

12. (a) Show that the function $f(x, y) = x + y$ is primitive recursive function. Hence compute the value of $f(2, 4)$. (10)

(b) Show that $f(x, y) = x^4$ is a primitive recursive function. (5)

13. (a) If H is a subgroup of index 2 in a group G then 'H' is a normal subgroup of G. (8)

(b) For any two subgroups H and K of a group G following hold (7)

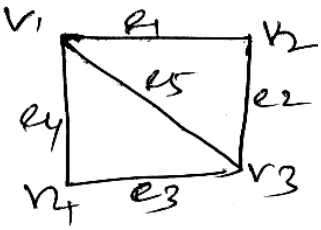
(i) $H \cap K$ is a subgroup of G

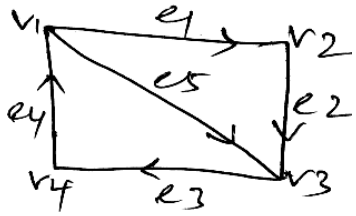
(ii) If H is normal in G then $H \cap K$ is normal in K.

14. (a) Prove that the order of each subgroup of a finite group G is a divisor of the order of the group G. (12)

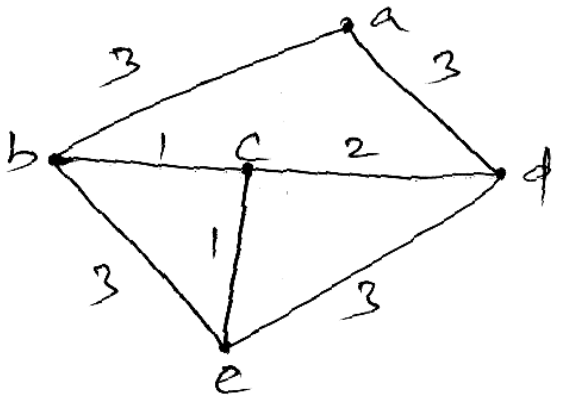
(b) Show that every cyclic group is an abelian group. (3)

15. (a) Find the incidence matrix to represent the graph shown below : (5)

(i)


(ii)


(b) Show how Kruskal's algorithm find a minimal spanning tree for the following graph. (10)



DE-690

Sub. Code
302

DISTANCE EDUCATION

M.C.A./ M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

COMPUTER NETWORKS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Mention the uses of Computer Networks.

2. What are the issues that occur in designing data link layer?

3. Briefly discuss on Routing and Switching.

4. Write a note on transport service primitives.

5. Give the importance of Data Compression?

6. How crash recovery is implemented in transport layer?

7. What is meant by Fragmentation?

8. List the benefits of broadband ISDN.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the Telephone System in detail.

10. Explain about the Sliding Window Protocols in detail.

11. Explain the Internetworking models.

12. Write short note on :

    (a) Wireless TCP and UDP

    (b) Write the services provided by transport layers to the upper layers.

13. Explain the Secret and Public Key algorithms.

14. Explain the Error detection and correcting codes.

15. Explain the tools in Multimedia.

DE–691

Sub. Code
303

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

SOFTWARE ENGINEERING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Define the terms software and software engineering. List the characteristics and components of software.

2. List the role of management in software development.

3. Define an estimation. Explain the LOC and FP estimations.

4. Write a short note on : software risk management.

5. Explain the software requirement specification in detail.

6. Describe the Jackson system development.

7. Explain the cyclomatic complexity in detail.

8. Briefly describe the black box testing.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Discuss the spiral model for software development.

10. Explain the various metrics categories in detail.

11. Describe the various resources required for software development.

12. Define the requirement analysis. List the requirement analysis tasks.

13. Define the CFD. Explain the CFD with an example.

14. Explain the various software design types in detail.

15. What is software testing? Explain the following :

    (a) Integration testing

    (b) Unit testing.

DE–692

Sub. Code
304

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

VISUAL PROGRAMMING

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly explain the menus in visual basic.

2. Discuss the different dialogs in visual basic.

3. Explain any five inbuilt functions in visual basic.

4. Describe the method to print the information in visual basic.

5. Illustrate the concept of class and object with an example program in VC++.

6. What are the resources available in VC++? Explain.

7. Write down the concept of serialization in VC++.

8. Enumerate the user interface classes in VC++.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. How to create and use controls in visual basic? Explain in detail.

10. Explain in detail about file system controls in visual basic.

11. Discuss the procedure to access database with the data control in visual basic.

12. Describe the procedure to import VBX controls in VC++.

13. Compare SDI and MDI environments in VC++ with neat diagram.

14. Write down in detail about the object linking and embedding in VC++.

15. Explain the procedure to create user defined DLL's in visual basic.

DE–9550

Sub. Code
305

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Third Semester

NETWORK LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the student by the lot system.

Each subdivision carries 50 marks.

1. (a) Write a Java program to add two numbers using RMI client server technique.

   (b) Write a Java program to find IP address of Local Host.

---------- Cut here ----------

2. (a) Write a Java program to find factorial of given number using RMI client, server technique.

   (b) Write a Java program to ENCRYPT a given word.

---------- Cut here ----------

3. (a) Write a Java program to implement address resolution protocol.

   (b) Write a Java program to test remote IP address for loop back address and multicast address.

4. (a) Write a Java program to set up a chat server and chat client.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

5. (a) Write a Java program to setup echo server and echo client.

   (b) Write a Java program to read a file from the remote system.

---------- Cut here ----------

6. (a) Write a Java program to send and receive message between client and server using UDP.

   (b) Write a Java program to display the user and their IP address logged in the network.

---------- Cut here ----------

7. (a) Write a Java program to subtract two numbers using RMI client server technique.

   (b) Write a Java program to send the following message: "EXECUTION IS SUCCESSFUL" from one system to another system.

---------- Cut here ----------

8. (a) Write a Java program to manipulate the IP address.

   (b) Write a Java program to create sockets for handing multiple connection.

DE–9551

Sub. Code
306

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2017.

Third Semester

VISUAL PROGRAMMING LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by the lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

---------- Cut here ----------

1. (a) Write a VB project that receives a year A from a text box and month as name from list box and displays number of days in the given month. Take care of leap years. Use Lost Focus event for list box.

   (b) Write a Visual C++ win32 Application program using MFC to create a new font.

2. (a) Write a VB project that stores 10 employee records with fields EMPNO, NAME, AGE, SEX, and SALARY, in an array. Display data in text boxes and provide command buttons to move to desired record.

(b) Write a Visual C++ win32 Application program to display "Good morning!" wherever the user clicks the mouse button on the client area.

---------- Cut here ----------

3. (a) Write a VB project that receives a foreign currency value selected from a list box and converts it into equivalent Indian rupees. (e.g USD 42.45, Sterling 71.30, D.Mark 25.52, SW Franc 31.58, SaudiRiyal 11.40, French Franc 7.60, UAE Dhiram 11.55, Kuwait Dhinar 140.56)

(b) Write a Visual C++ win32 Application program using MFC to create a list box for displaying the name of the states in India.

---------- Cut here ----------

4. (a) Write a VB project using control array to create a scientific calculator with appropriate command buttons. Include the following capabilities for the calculator : +, –, *, /, %, power, square root, square and log(base 10).

(b) Write a Visual C++ win32 Application program using MFC to fill the background of the client area with bitmap image.

5. (a) Write a VB project for commercial bank operations using Savings Bank account database, with the following features :

   (i) ADD NEW ACCOUNT

   (ii) DEPOSIT AMOUNT

   (iii) WITHDRAW AMOUNT (with minimum balance condition)

   (b) Write a Visual C++ win32 Application program using MFC to display the status of ALT, CTRL, SHIFT, NUM LOCK and SCROLL LOCK keys.

---------- Cut here ----------

6. (a) Write a VB project using built in ActiveX control (Rich Text Box) to develop a windows NOTEPAD like editor with File and Edit menus with the popup menu whenever necessary.

   (b) Write a Visual C++ win32 Application program using MFC to display the current mouse coordinates in the status bar.

---------- Cut here ----------

7. (a) Write a VB project using ActiveX DLL or EXE to add a class module to perform the following functions :

   (i) Test whether the given number is perfect or not

(ii) Whether the given number is Armstrong or not

(iii) Find the factorial of the given number

(iv) Sum of digits

(b) Write a Visual C++ win32 Application program using MFC to create two push buttons, OK and CANCEL on the client area. Buttons should respond to user click over them for displaying appropriate messages.

DE–693

Sub. Code
401

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Briefly explain various concepts of accounting.

2. From following Balance Sheets prepare a statement of changes in working capital.

Balance Sheets

Liabilities	2005 Rs.	2006 Rs.	Assets	2005 Rs.	2006 Rs.
Equity capital	5,50,000	5,50,000	Fixed assets	6,00,000	7,00,000
Debentures	3,70,000	4,50,000	Long-term investments	2,00,000	1,00,000
Tax payable	77,000	43,000	Work-in-progress	80,000	90,000
Creditors	96,000	1,92,000	Stock	1,50,000	2,25,000
Bills payable	37,000	30,000	Debtors	70,000	1,40,000
			Cash	30,000	10,000
	11,30,000	12,65,000		11,30,000	12,65,000

3. Calculate the material cost variance, material price variance and material usage variance from the following :

	Standard			Actual		
	Quantity	Unit price	Total	Quantity	Unit price	Total
Material A	100	20	200	50	30	150
Material B	200	30	600	100	60	600
Material C	200	60	1200	150	50	750

4. Prepare a production budget for three months ending March 31, 2009, for a factory producing four products, on the basis on the following information.

Type of product	Estimated stock on 1st, Jan, 2015	Estimated sales during Jan – March 2015	Estimate stock on Mar, 31, 2015
A	2,000	10,000	3,000
B	3,000	15,000	5,000
C	4,000	13,000	3,000
D	3,000	12,000	2,000

5. From the following particulars prepare cost sheet.

	Rs.
Raw materials	1,00,000
Direct wages	40,000
Factory overheads	30,000
Administrative overheads	25,000
Selling and distribution overheads	30,000

6. From the following Profit and Loss Account of X Ltd. for the year ending 31st March, 2007 you are required to calculate (a) Gross profit ratio (b) Net profit ratio (c) Operating profit ratio and (d) Operating ratio.

Profit and Loss Account

Particulars	Rs.	Particulars	Rs.
To Opening Stock	2,50,000	By Sales	18,00,000
To Purchase	10,50,000	By Closing Stock	1,50,000
To Gross Profit	6,50,000		
	19,50,000		19,50,000
To Selling and Distribution expenses	1,00,000	By Gross Profit	6,50,000
To Administration Expenses	2,30,000	By Profit on sale of fixed assets	50,000
To Finance Expenses	20,000		
To Net profit	3,50,000		
	7,00,000		7,00,000

7. Write a note on Time Value of Money and its significance.

8. Explain about zero base budgeting.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. From the following information you are required to prepare a Balance Sheet :

(a) Current Ratio – 1.75

(b) Liquid Ratio – 1.25

(c) Stock Turnover Ratio (Cost of Sales/Closing Stock) – 9

(d) Gross profit Ratio – 25%

(e) Debt collection period – $1^1/_2$ months

(f) Reserves and Surplus to Capital – 0.2

(g) Turnover of Fixed Assets – 1.2

(h) Capital Gearing Ratio – 0.6

(i) Fixed Assets to Net worth – 1.25

(j) Sales for the year Rs. 12,00,000.

10. The following is the trial balance of Mr. Kumar as on 31st March, 2008 :

	Debit Rs.	Credit Rs.
Sales		1,20,000
Purchases	90,000	
Wages	5,000	
Opening stock	12,000	
Sundry Debtors	22,000	
Sundry Creditors		11,000
Bills receivable	10,000	
Bills payable		8,000
Bad debts	2,000	
Carriage inwards	1,000	
Telephone charges	2,000	
Printing & stationery	500	
Investments	6,000	
Interest		800
Land & buildings	40,000	
Rent		3,200
Drawings	3,000	
Cash	1,500	
Furniture	5,000	
Capital		57,000
	2,00,000	2,00,000

Adjustments

(a) Closing stock is valued at Rs. 22,000.

(b) Wages outstanding Rs. 300.

(c) Interest receivable amounts to Rs. 200.

(d) Rent received in advance amounts to Rs. 400.

(e) Depreciation @ 10% is required on furniture.

Prepare Final accounts.

11. Assuming that the cost structure and selling prices remain the same in periods I and II find out :

(a) P/V Ratio

(b) B.E. sales

(c) Profit when sales are Rs. 1,00,000

(d) Sales required to earn a profit of Rs. 20,000

(e) Margin of safety in II period.

Period	Sales Rs.	Profit Rs.
I	1,20,000	9,000
II	1,40,000	13,000

12. Prepare a flexible budget for overheads on the basis of the following data and ascertain overhead rates at 50%, 60% and 70% capacity.

	At 60% capacity Rs.
Variable overheads	
Indirect material	6,000
Indirect labour	18,000
Semi-variable overheads	
Electricity (40% fixed 60% variable)	30,000
Repairs (80% fixed 20% variable)	3,000
Fixed overheads	
Depreciation	16,500
Insurance	4,500
Salaries	15,000
Total overheads	93,000
Estimated direct labour hours	1,86,000

13. Explain the objectives of financial management.

14. The X ltd. is considering the purchase of a new machine. Two alternative machines (A and B) have been suggested, each having an initial cost or Rs. 4,00,000. Earnings after taxation are expected and present value at 10% as follows.

Year	Cash inflows		Present value at 10%
	Machine A	Machine B	
1	40,000	1,20,000	0.91
2	1,20,000	1,60,000	0.83
3	1,60,000	2,00,000	0.75
4	2,40,000	1,20,000	0.68
5	1,60,000	80,000	0.62

Evaluate and choose one of the machines under net present value method.

15. What is break even analysis? Enumerate its merits and demerits.

DE–694

Sub. Code
402

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2017.

Fourth Semester

INTERNET PROGRAMMING

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Give an overview on internet programming.

2. What are the design issues of WWW? Explain.

3. State the features of java script.

4. Write a simple program to illustrate the use of VB script.

5. Compare HTML and SGML.

6. Create an Active-X control to activate a web page.

7. Mention the procedure to add graphics to web pages with an example.

8. Specify the steps to create Netscape navigator plug-ins.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. How Java is useful in doing internet programming? Explain.

10. Discuss in detail about CGI with respect to internet programming.

11. Describe the scripting languages other than VB script and java script.

12. Explain in detail about Shokwave and Lingo.

13. Write in detail about HTML tags with an example for each.

14. Write down the steps for creating a custom integrated application with multiple protocols.

15. How to design a web page and website with images and hyperlinks? Explain in detail.

DE–695

Sub. Code
403

DISTANCE EDUCATION

M.C.A./M.C.A. (LE) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions

1. Illustrate the merits and demerits of modeling a System.

2. Explain the relationships among various Classes of a System with an example.

3. Explain the issues in identifying the proper Classes and Objects in a System.

4. Draw a Use Case diagram for the functions of a Banking System.

5. List various components of Class diagram.

6. Draw a Package diagram for the Air ticket Reservation system.

7. What we can observe from a Deployment diagram of a system? Explain with an example.

8. Discuss the strategies used to identify the conceptual classes.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Write a detailed note on

   (a) Class

   (b) Objects

   (c) Model.

10. Describe the Key abstraction mechanism.

11. Discuss the relationships among various Objects of a System.

12. Describe the various activities of the View process.

13. Sketch a Class diagram for the functions of a College.

14. Draw an Interactive diagram for Banking System.

15. Draw a State diagram for withdrawing money from an ATM.

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DE–696

Sub. Code
404

DISTANCE EDUCATION

M.C.A./M.C.A. (LE) DEGREE EXAMINATION, MAY 2017.

Fourth Semester

COMPILER DESIGN

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions

1. Write a note on finite Automata.
2. How regular expressions are created?
3. What is meant by Top Down parsing? Explain.
4. Write a note on Dead Code Elimination.
5. Explain Data Structure for Symbol Table.
6. Explain the importance of flow graphs.
7. Explain the concept of Code Propagation.
8. What is meant by global Optimization? Explain.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. (a) What are the issues in Lexical Analysis? (9)

   (b) Elaborate the recognition of tokens. (6)

10. (a) Explain Bootstrapping. (10)

(b) Distinguish between phases and passes. (5)

11. Construct the predictive parser for the following grammar.

S-> (L)/a

L->L,S/S

12. (a) Explain in detail about the specification of a simple type checker. (6)

(b) How to subdivide a run-time memory into code and data areas? Explain. (9)

13. Construct SLR parsing table for the following grammar.

E $\rightarrow$ E+T/J

T $\rightarrow$ T*F/F

F $\rightarrow$ (E)/id.

14. (a) Explain in detail about the various issues in design of code generator. (9)

(b) Write an algorithm to partition a sequence of three address - Statements into basis blocks. (6)

15. (a) Explain the principle sources of optimization in detail. (5)

(b) Discuss the various peephole optimization techniques in detail. (10)

DE–9552

Sub. Code
405

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2017.

Fourth Semester

INTERNET PROGRAMMING LAB

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose ONE question to each candidate by lot system.

Each sub division carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging and Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a Java script program to create a color palette and change the background and foreground colors.

   (b) Write a Java program to create database application for adding name with phone number and display it using MS-Access.

2. (a) Write a Java script program to create an order form to select the house articles.

   (b) Write a Java program to create database application for implementing student mark list using MS-Access.

---------- Cut here ----------

3. (a) Write a VBscript script program to validate username and password.

   (b) Write an applet program for displaying different images using border layout.

---------- Cut here ----------

4. (a) Write Swing program to create button with images.

   (b) Write an Html program to create a screen with different images, colors and table.

---------- Cut here ----------

5. (a) Create a simple web page for School Management System.

   (b) Write an applet program to work with different colors and fonts.

DE–9553

Sub. Code
406

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2017.

Fourth Semester

COMPILERN DESIGN – LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

ONE question is to be given to each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. Consider the following regular expressions:

   (a) $(0 + l)^* + 0^*1^*$

   (b) $(ab^*c + (def)^+ + a^*d^+e)^+$

   (c) $((a + b)^*(c + d)^*)^+ + ab^*c^*d$

   Write separate programs for each of the regular expressions mentioned above.

---------- Cut here ----------

2. Design a Lexical analyzer for identifying different types of token used in C language.

3. Given a text-file which contains some regular expressions, with only one RE in each line of the file. Write a program which accepts a string from the user and reports which regular expression accepts that string. If no RE from the file accepts the string, then report that no RE is matched.

---------- Cut here ----------

4. Write a program to eliminate left recursion.

---------- Cut here ----------

5. Write a program for Recursive Descent Calculator.

---------- Cut here ----------

6. Write a program that recognizes different types of English words.

---------- Cut here ----------

7. Consider the following grammar:

$S \rightarrow ABC$

$A \rightarrow abA \mid ab$

$B \rightarrow b \mid BC$

$C \rightarrow c \mid cC$

Following any suitable parsing technique (prefer top-down), design a parser which accepts a string and tells whether the string is accepted by above grammar or not.

8. Write a program which reads a left-recursive regular grammar and removes left recursion from the grammar.

   For example: A possible input grammar is

   $S \rightarrow Sa$

   $S \rightarrow a$

   Its output after removal of left-recursion will be

   $S \rightarrow aS$

   $S \rightarrow a$

---------- Cut here ----------

9. Write a program which accepts a regular grammar with no left-recursion and no null-production rules, and then it accepts a string and reports whether the string is accepted by the grammar or not.

---------- Cut here ----------

10. Design a parser which accepts a mathematical expression (containing Integers only). If the expression is valid, then evaluate the expression else report that the expression is invalid.

DE–697

Sub. Code
501

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

DISTRIBUTED COMPUTING

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write a brief note on the Mobile and Ubiquitous computing.

2. Explain the different switching schemes used in a network.

3. Explain the role of XML in interprocess communication.

4. Discuss the design issues for RPC.

5. Explain the file service architecture for distributed systems.

6. Explain the different names and its attributes used in distributed file service.

7. Write a brief note on network time protocol.

8. Explain how nested transaction is handled in distributed system.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. What is Architectural model? Explain in detail the architectural element and architectural Pattern.

10. Explain the APIs used for interprocess communication.

11. Explain the Request Reply protocol.

12. Outline an implementation for the Election service that ensures that its records remain consistent when it is accessed concurrently by multiple clients.

13. Explain the different security techniques and mechanisms used for securing a distributed system.

14. Explain the use of digital signature in securing a system.

15. What is a deadlock in distributed system? Explain how deadlocks can be detected and resolved in distributed systems?

DE–698

Sub. Code
502

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

WEB TECHNOLOGY

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain about <FORM> and <FRAME> tags in detail.

2. What is XML? Explain the use of XML Namespace?

3. List the advantages of Java Beans.

4. What is EJB? Explain in detail.

5. Explain Lifecycle of a Servlet.

6. How does JSP work? List the advantages of servlet over JSP.

7. List the JSP implicit objects and explain.

8. Short notes on Struts framework.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the concept of Image maps by means of HTML with an Example.

10. Explain in detail the various methods of DOM and SAX.

11. What is Java Beans? Explain Bound properties with suitable example.

12. Explain the following :

    (a) Handling HTTP request and responses.

    (b) Cookies - session tracking.

    (c) Security issues.

13. Explain the steps to install Tomcat Server and its testing environment.

14. Explain in detailed about JSP Application Development.

15. Explain how to access a database from JSP Page with a suitable example.

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DE-699

Sub. Code
503

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

.NET FRAME WORKS

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare .NET with other programming languages.
2. Briefly explain .NET Building Block Services.
3. What is Assembly? Explain the different types of Assembles.
4. Explain about CTS and CLS.
5. Explain about attributes and its types with an example.
6. Explain about Relational database and SQL in accessing data with ADO.NET.
7. What are the XML documentation comments? Explain with simple program.
8. Briefly explain about WSDL.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain the Architecture of .NET Framework.
10. Explain about the activity of Common Language Runtime.

11. Discuss in detailed about Memory management in CLR.

12. Explain about various String Handling methods in C# with suitable example

13. Explain how to build a Client and Server in .NET Framework.

14. Write a program in any .NET language for scheduling threads.

15. Define ADO .NET? Explain how to access data with ADO .NET.

DE-700

Sub. Code
504

DISTANCE EDUCATION

M.C.A./M.C.A. (L) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

DATA MINING AND WAREHOUSING

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE of the following questions.

1. Write a short note on Self Learning Computer System

2. Write a short note on Knowledge Discovery Process and Data Selection.

3. Briefly describe about Genetic Algorithms in Data Mining Process.

4. What are the steps involved for the Identification of Facts and Dimensions.

5. Explain about the Vertical Partitioning.

6. Write a short note on Disaster Recovery.

7. What is the Process involved during Capacity Planning.

8. Write a short note on testing the database.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR of the following questions.

9. Explain in detail about Data Mining System Architecture and its issues and Challenges.

10. Write a short note on :

    (a) Data Selection

    (b) Cleaning

    (c) Enrichment.

11. Explain in detail about the Typical Process Flow within a Data Warehouse.

12. Explain in detail about how overnight process helps to operate the Data Warehouse.

13. Describe in detail about Service Level Agreement (SLA).

14. Explain detail about the following :

    (a) Data Mining Vs Query Tools.

    (b) Data Mining needs from Data Warehouse.

15. Explain :

    (a) Testing the operational environment

    (b) Testing the Application

    (c) Logistics of the Test.

DE–9554

Sub. Code
505

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017

Fifth Semester

.NET LAB

(2010 Academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a VB.Net program to accept a string and convert the case of the characters.

   (b) Create table called employee with the fields employee id, name, age, sex, address mail id and then insert few customer information into customer table.

---------- Cut here ----------

2. (a) Develop a VB.Net application using Datagrid to display records.

   (b) Create a currency converter applications in ASP.NET with two input text boxes named US dollar and Indian rupees and a submit button named convert.

3. (a) Write a VB .Net Program to bind data using Hyperlink column in data grid.

   (b) Create an application that displays the information of all students without selection, edition and sorting options.

---------- Cut here ----------

4. (a) Design a form to select image from list and display it in the picture box.

   (b) Develop an Employee payroll system to generate a crystal report from an existing database using VB.NET.

---------- Cut here ----------

5. (a) Write a VB. Net program to display the Holiday in calendar.

   (b) Write an ASP.NET program that gets user input such as the user name, mode of payment, appropriate credit card. After the user enters the appropriate values the validation button must validates the values entered.

---------- Cut here ----------

6. (a) Develop a database application to store the details of students using ADO.NET

   (b) Write a Program in ASP that has a form taking the users name as input. Store this name in a permanent cookie & whenever the page is opened again, then value of the name field should be attached with the cookies content.

7. (a) Develop a web service to fetch data from a table and send it across to the client.

(b) Write a Program to display the difference between the two dates in the calendar

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DE–9555

Sub. Code
506

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Fifth Semester

WEB TECHNOLOGY LAB

(2010 Academic Year Onwards)

Time : 3 Hours Maximum : 100 Marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Create an HTML table with five rows and seven columns and split them using Rowspan and Colspan with your Own Records.

   (b) Create a Home page for your college information using Text links, align the text with colors.

---------- Cut here ----------

2. (a) Write an XML document to display Employee profile.

   (b) Create a web page for library Home page. Align the text with colors. Use the links with various pages.

---------- Cut here ----------

3. (a) Write a JSP program to prepare employee pay bill using JDBC.

   (b) Write an ASP program to prepare stock details in super market using java script.

4. (a) Write a servlet program to display the request and response header information.

   (b) Create an application form for MCA Course using various HTML controls.

---------- Cut here ----------

5. (a) Create a HTML frame using various controls.

   (b) Write a servlet program to create a chatting application.

---------- Cut here ----------

6. (a) Write an XML document to display your favourite Indian cricket players. Write an XSL style sheet and attach that to the XML document validate the document using DTD or XSD.

   (b) Create an HTML table for your Class time table using Row span and Col span.

---------- Cut here ----------

7. (a) Write an XML document to display Student resume.

   (b) Write a HTML program to pick the color from color palette and display color name in text box.

---------- Cut here ----------

8. (a) Write a Servlet program to display "Hello world".

   (b) Create a Image animation using Java script.

DE–701

Sub. Code
601

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

MOBILE COMMUNICATIONS

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Write short notes on Localization.

2. Compare FDMA and CDMA.

3. Brief describe about Bluetooth.

4. Differentiate Infra-red against Radio Transmission.

5. Explain how optimization technique used in Mobile IP.

6. Describe Congestion Control Mechanism in Traditional TCP.

7. Write a note on slow start mechanism.

8. What are the issues faced during Data Transmission?

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain in detail about GSM Architecture.

10. Discuss about IEEE802.11 Standards with neat diagram.

11. Explain in detail about Ad-hoc Networks with some basic protocol used in these networks.

12. Write a short notes on

    (a) DHCP (Dynamic Host Configuration Protocol).

    (b) Reverse Tunneling.

13. Explain the following in Indirect TCP.

    (a) Fast retransmit/Fast Recovery.

    (b) Transmission/ Time-out freezing.

14. Explain with Importance, advantage and disadvantages for the following:

    (a) Selective retransmission

    (b) Transaction oriented TCP with its connection setup.

15. Discuss about WAP and explain the importance of components that are interface with WAP Architecture.

DE–702

Sub. Code
602

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

MIDDLEWARE TECHNOLOGY

(2010 Academic Year onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Compare Two-tier model with Three-tier model.

2. Discuss about DOTNET Architecture and its importance.

3. With an example explain how XML differ with other markup language.

4. Write a note on Entity Bean.

5. List out Various HTML Controls.

6. Discuss about security management.

7. Write about XML namespace.

8. Differentiate Provider objects with Consumer objects.

SECTION B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Explain about the importance of MVC Architecture used in middleware technology.

10. Discuss in detail about JDBC Services.

11. Explain in detail about Session Beans with examples.

12. With an example explain the following controls

    (a) Ad Rotator Controls

    (b) Calendar Controls.

13. Write the Rules of parser in ASP.NET with suitable example.

14. Developing a simple web service application for the addition of two numbers and getting weather report of particular city using Post method.

15. Explain the following Web Services

    (a) UDDI

    (b) SOAP

    (c) WSDL.

DE–703

Sub. Code
603

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2017.

Sixth Semester

AGENT BASED INTELLIGENT SYSTEM

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Give a brief note on Rational Agents.

2. What is the need for optimal decision in games? Explain.

3. Write short notes on Resolution Strategies.

4. Explain the State- Space search.

5. Give a brief note on Conditional Planning.

6. Explain the Utility Theorem.

7. Bring out the significance of Knowledge in Learning.

8. Give a brief note on Reinforcement Learning.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. Give a detailed description on agents and the nature of environments.

10. Illustrate the concept of problem solving with an example.

11. Describe the syntax and semantics of First-order logic.

12. Give a detailed note on forward chaining.

13. Explain the planning problem.

14. Give a detailed description on Bayesian networks.

15. Bring out the significance of communication for agents.

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DE–704

Sub. Code
604

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

OPEN SOURCE ARCHITECTURE

(2010 Academic Year Onwards)

Time : Three hours Maximum : 100 marks

PART A — (5 × 8 = 40 marks)

Answer any FIVE questions.

1. Explain the advantages of open source software.

2. Explain the basics of TCP/IP networking.

3. What is the procedure to setting up web servers in X windows system?

4. Explain in detail about GNU debugging tools.

5. Write short notes on autoconf and autogen.

6. Explain the source code management using CVS.

7. Write short notes on GTK programming.

8. Explain the features of Python language.

PART B — (4 × 15 = 60 marks)

Answer any FOUR questions.

9. How to install linux operating system in bash? Illustrate with an example.

10. Explain the usage of netfilter and iptables with examples.

11. Explain the different types of routers with examples.

12. Write short notes on patch and diff with examples.

13. Explain the architecture of X windows server Architecture.

14. Discuss the data types and identifiers in python with examples.

15. Explain the control statements in Python with examples.

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DE–9556

Sub. Code
605

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) (Semester) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

OPEN SOURCE PROGRAMMING LAB

MCA(s) (2010 Academic Year Onwards)/MCA(s) Lateral

Time : 3 Hours Maximum : 100 Marks

Examiner should select and give ONE question to each candidate by lot system.

1. (a) Write a PHP Program to reserve a train ticket using MySQL.

   (b) Write a PHP Program to create a web page for online shopping.

---------- Cut here ----------

2. (a) Write a PHP Program to retrieve the employee details starting with Alphabets with 'B' from top 3 salary using MySQL.

   (b) Write a PHP Program to create a directory and to view the contents of directory.

---------- Cut here ----------

3. (a) Write a PHP Program to create a curriculum viate using PHP and HTML.

   (b) Write a Program a simple online Tax payment using PHP and HTML.

4. (a) Write a PHP Program to create a Employee table in MYSQL.

   (b) Write a simple PHP Program to check that emails are valid.

---------- Cut here ----------

5. (a) Write a PHP script to check whether the page is called from https or http.

   (b) Write a PHP program to insert and select data into students database using MYSQL.

---------- Cut here ----------

6. (a) Write a PHP Program to update the transaction amount after debit the amount to another account using MYSQL.

   (b) Write a PHP Program to count number of lines in a file.

---------- Cut here ----------

7. (a) Write a PHP Program to validate user name with security code to allow transaction using MYSQL.

   (b) Write a simple Program to select the image as your profile picture for your web page using PHP and HTML.

---------- Cut here ----------

8. (a) Write a PHP Program to check the availability of seats in flight from particular source to destination using MYSQL.

   (b) Write a PHP Program for the following

      (i) Reverse a string

      (ii) Concatenation of two strings.

DE–9557

Sub. Code
606

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, MAY 2017.

Sixth Semester

MULTIMEDIA TOOLS LAB

(2010 academic year onwards)

Time : Three hours Maximum : 100 marks

Examiner has to choose any ONE question from each candidate by lot system.

Break-Up of Marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Execution	:	35
Result	:	10
Total	:	100

1. (a) Develop an animation in Flash to show a falling tree.

   (b) Implement a program in FLASH to develop a bouncing ball Animation.

2. (a) Develop an animation in Flash to create a blinking eye.

   (b) Implement a program to animate the flight in a runway.

---------- Cut here ----------

3. (a) Develop an animation in flash to demonstrate Masking effect.

   (b) Develop an animation using FLASH to create a rainfall effect.

---------- Cut here ----------

4. (a) Implement a animation using FLASH with a button for making five faces and change them automatically.

   (b) Implement an animation to show a flying airplane.

---------- Cut here ----------

5. (a) Create any two different shapes animate the shapes slowly appear and disappear one after another.

   (b) Animate your own text to create the horizontal flipping effect.

---------- Cut here ----------

6. (a) Develop an animation in FLASH to glow a bulb with changing colours.

   (b) Implement a program to create a Maze using Maya.

---------- Cut here ----------

7. (a) Implement a program in flash forgetting rain fall effect.

   (b) Implement a program for animating a flight hitting the twin tower using flash.

8. (a) Create a cartoon with Laughing face using FLASH.

   (b) Implement a program in FLASH for bouncing a ball with a pad moving along with the ball.

---------- Cut here ----------

9. (a) Animate your name with different colours. Make the characters to blink.

   (b) Develop an animation in flash to show a moving car in the road.

---------- Cut here ----------

10. (a) Develop an animation to show a flying butterfly in a garden.

    (b) Develop an animation to show a brick falling to ground from a shelf.